

EXP No:1A	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK
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AIM:

To configure Python, Jupyter Notebook and verify the installation of essential data analytics libraries using the Baseball Databank dataset

PROCEDURE:

1. Install Jupyter Notebook using the command `pip install notebook`.
2. Create a new Python notebook and rename it appropriately.
3. Import the required libraries such as NumPy, Pandas, and Matplotlib.
4. Verify the successful installation of the libraries.
5. Load the Baseball Databank dataset (Batting.csv) using the Pandas `read_csv()` function.
6. Display the first few records of the dataset to verify successful loading.
7. Examine the dataset structure using functions such as `shape` and `head()`.
8. Perform basic NumPy operations on the runs data.
9. Generate a line graph using Matplotlib to visualize total runs by year.

OUTPUT:

```
In [1]: print("Welcome to Baseball Data Analytics Lab")
```

Welcome to Baseball Data Analytics Lab

```
In [2]: a = 20
b = 10

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

```
Out[2]: Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2.0
```

```
In [3]: dataset_name = "Baseball Databank"
sport = "Baseball"
competition = "Major League Baseball"

print("Dataset:", dataset_name)
print("Sport:", sport)
print("Competition:", competition)
```

```
Out[3]: Dataset: Baseball Databank
Sport: Baseball
Competition: Major League Baseball
```

```
In [4]: import numpy as np

runs = np.array([10, 20, 30, 40, 50])

print("Runs:", runs)
print("Mean Runs =", np.mean(runs))
print("Total Runs =", np.sum(runs))
```

```
Out[4]: Runs: [10 20 30 40 50]
Mean Runs = 30.0
Total Runs = 150
```

```
In [5]: import pandas as pd

def find_data(filename):
    candidates = [
        Path(filename),
        Path("data") / filename,
        Path("../data") / filename,
        Path("../..data") / filename,
        Path("/content") / filename,
    ]
    for p in candidates:
        if p.exists():
            return p
    raise FileNotFoundError(f"{filename} not found. Upload the Baseball Databank CSV or keep it in the repository data folder.")

import pandas as pd

df = pd.read_csv(find_data("Batting.csv"))

print("First 5 records of Baseball batting dataset:")
display(df.head())

print("Dataset Shape: ", df.shape)
print("Number of Columns: ", len(df.columns))
```

First 5 records of Baseball batting dataset:

	playerID	yearID	stint	teamID	lgID	G	AB	R	H	2B	3B	HR	RBI	SB	CS	BB	SO	IBB	HBP	SH	SF	GIDP
0	10001	2000	1	NYN	AL	8	22	6	7	1	0	2	6	2	4	4	0	0	0	0	0	1
1	10002	2000	1	BOS	AL	6	15	2	4	0	0	0	0	2	1	1	4	2	0	0	0	0
2	10003	2000	1	CHN	NL	7	18	3	5	1	0	1	3	2	5	5	3	0	0	0	0	0
3	10004	2000	1	NYN	AL	6	20	4	6	2	0	1	4	0	1	3	3	0	0	0	0	1
4	10005	2000	1	SEA	AL	5	14	2	4	1	0	2	2	0	1	3	0	0	0	0	0	0

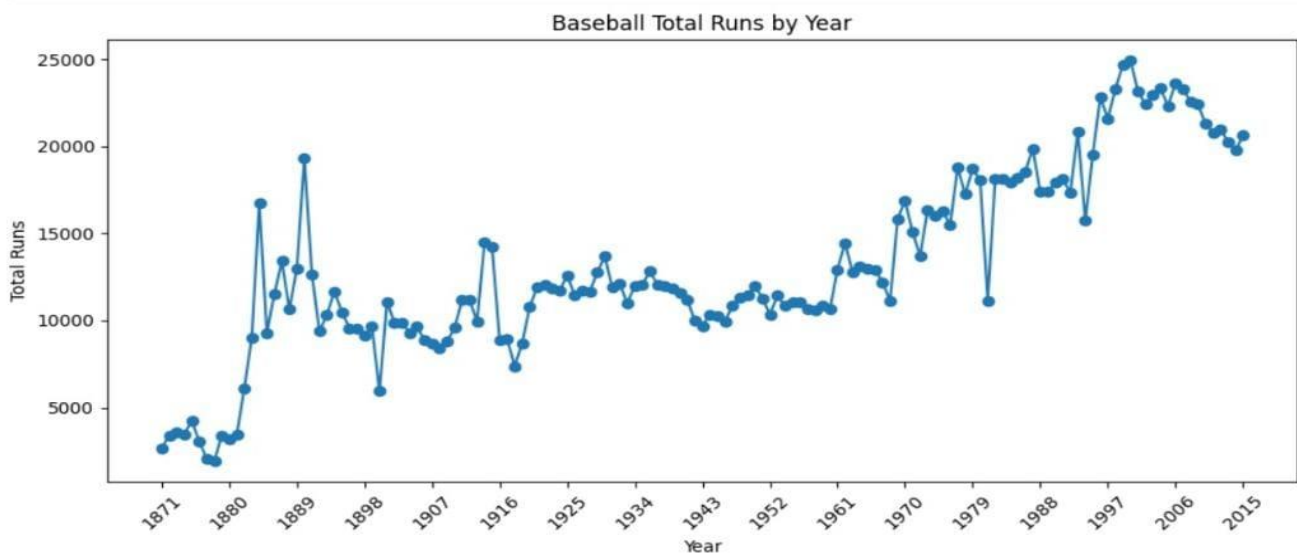
Dataset Shape: (32213, 22)

Number of Columns: 22

```
In [6]: import matplotlib.pyplot as plt

season_runs = df.groupby("yearID")["R"].sum()

plt.figure(figsize=(10, 5))
plt.plot(season_runs.index, season_runs.values, marker="o")
plt.title("Baseball Total Runs by Year")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.xticks(season_runs.index[::max(1, len(season_runs)//15)], rotation=45)
plt.tight_layout()
plt.show()
```



```
In [7]: print("NumPy imported successfully")
print("Pandas imported successfully")
print("Matplotlib imported successfully")
print("Baseball dataset loaded successfully")
```

```
NumPy imported successfully
Pandas imported successfully
Matplotlib imported successfully
Baseball dataset loaded successfully
```

RESULT:

Thus, the Python environment and Jupyter Notebook were configured successfully, and the essential data analytics libraries were verified using the Baseball dataset.